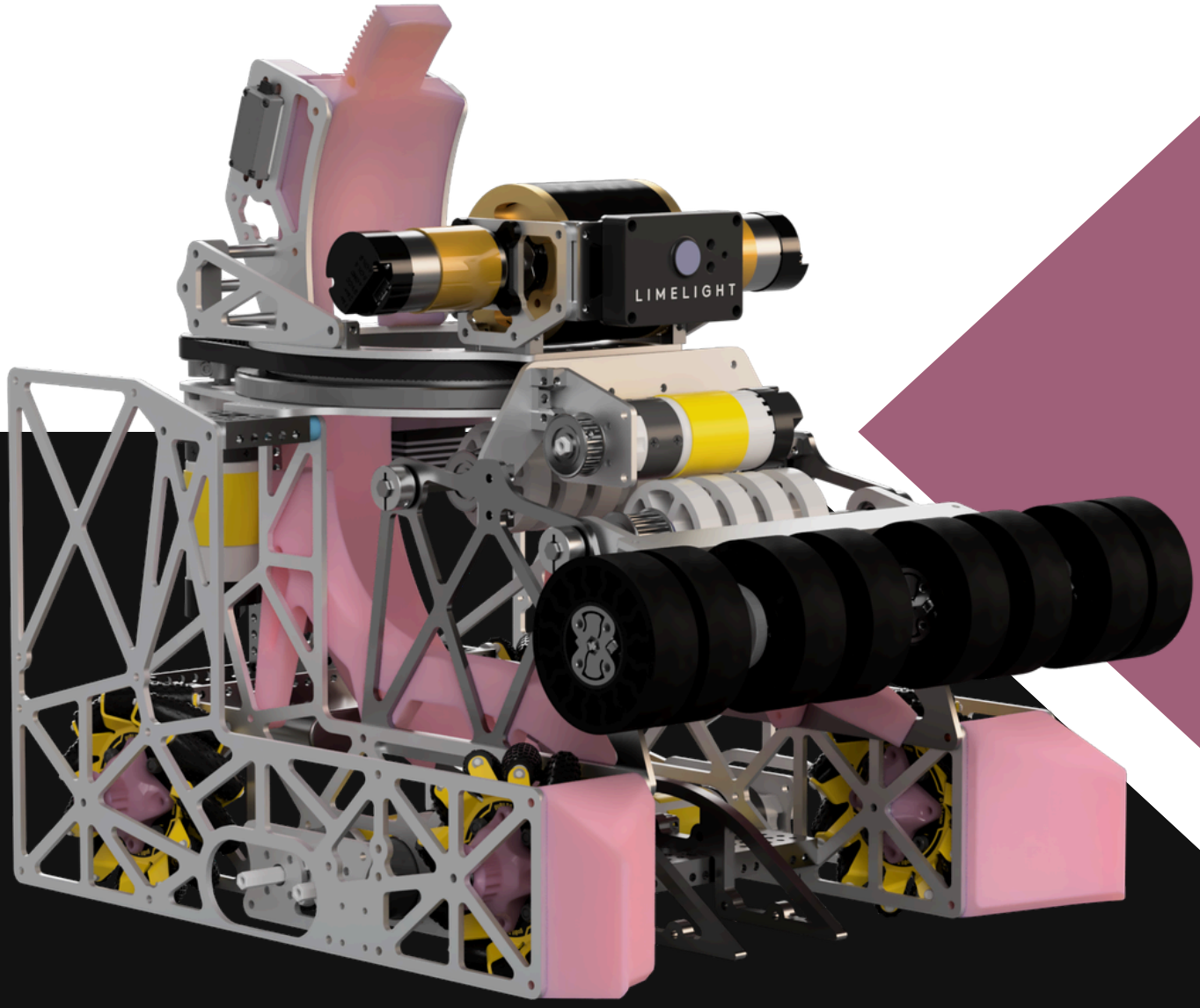
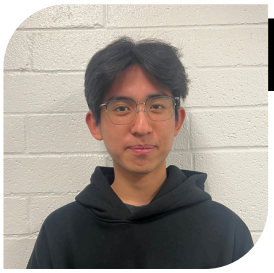
TEAM 359
WEBB.EXE

CHASING INNOVATION



WE ARE COMMITTED TO ENGINEERING INNOVATIVE, HIGH-PERFORMANCE ROBOTS BY COLLABORATING WITH PURPOSE, CULTIVATING A DYNAMIC, GROWTH-DRIVEN ENVIRONMENT, AND INSPIRING THE NEXT GENERATION OF ENGINEERS THROUGH FIRST.

MEET THE TEAM



JERRY H.

Captain
Mechanic Lead
3rd year FIRST

Skills Developed: Team management, collaborative designing
Team Highlight: Mentoring new members for the whole Webb Robotics Program

LINDA M.

Captain
Outreach lead
3rd year FIRST

Skills Developed: Team management, Outreach
Team Highlight: Leading one of our most successful outreach programs to date

ABIGAIL S.

Outreach Lead
2nd year FIRST

Skills Developed: Team management, Advanced CAD Design
Team Highlight: Actively developing leadership skills

TIMMY L.

Programming Lead
Team Treasurer
3rd year FIRST

Skills Developed: Team management, advanced automation
Team Highlight: Stepping up to Programming lead without a prior year in 359

ANNI H.

Mechanic
1st year FIRST

Skills Developed: Advanced CAD Design
Team Highlight: Self-taught CAD expertise and optimized intake

ETHAN C.

Outreach
1st year FTC, 4rd year FIRST

Skills Developed: Event Coordination, Communication
Team Highlight: Identified and contacted teams in need of mentoring

NORA M.

Programming
Outreach
1st year FIRST

Skills Developed: Limelight development
Team Highlight: Key role in intergrating our limelight auto tracking system

CLAIRE J.

Programming
Outreach
1st year FIRST

Skills Developed: Game Development, event coordination
Team Highlight: Led private python lessons for North Korean refugee students

RANIN A.

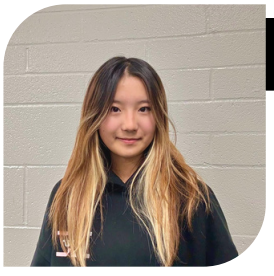
Programming
Outreach
1st year FIRST

Skills Developed: Game development and programming
Team Highlight: Led the partnership with the museum on the paleo game

OLIVIA K.

Mechanic
1st year FIRST

Skills Developed: Robot mechanism assembly
Team Highlight: Assembled and iterated on our robot's intake subsystem



ANNA X.

Social Media
Outreach
1st year FIRST

Skills Developed: Social media and communication with experts
Team Highlight: Connected us with experts and sponsors

KAYLEE

Mechanic
1st year FIRST

Skills Developed: Design Process
Team Highlight: Iterated on our robot hood mechanism

MEET THE TEAM

ABOUT US

We are a **19-year** FTC team, Team #359 "WEBB.EXE" from The Webb Schools in Claremont, California. Our mission is to **develop, inspire, and educate** by fostering a **culture** of innovation and a passion for learning within our community and beyond, throughout our participation in FTC. We strive to **empower** students from diverse backgrounds, promoting a prospering environment where every member can thrive and contribute.

ROBOT OVERVIEW

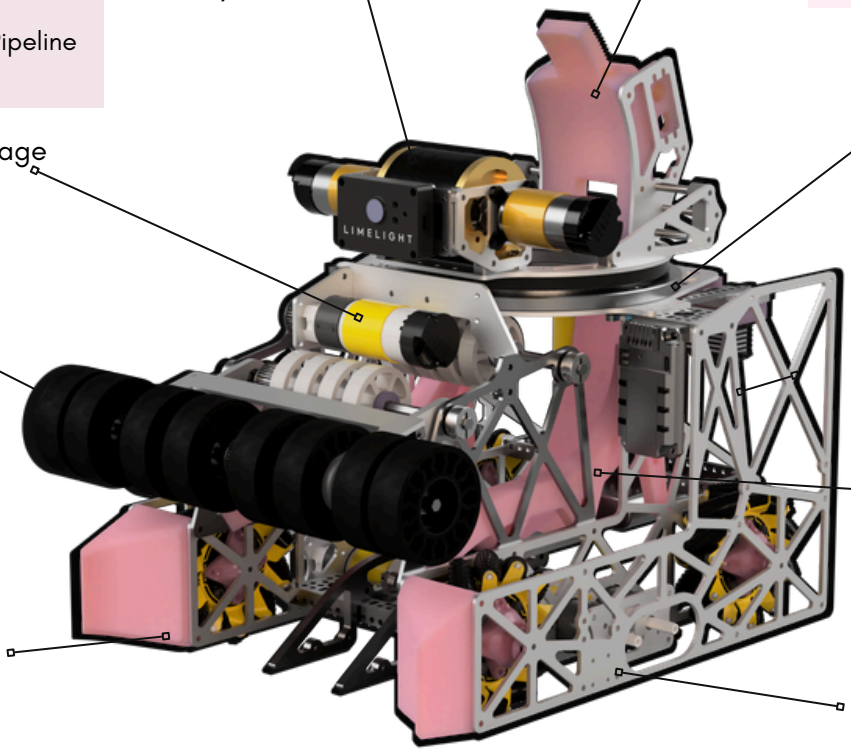
Intelligent Control

Multi-threaded asynchronous teleop, enabling:

- Accurate input response
- Asynchronous command timing
- Individual Vision/Tracking Pipeline for consistency

Custom Parts

- 95% custom parts
- 3dprinting, laser cutting, plasma cutting, and CNC
- Functional and aesthetic designs



Autonomous

- Multithreaded Autonomous Pathing Library
- Pedro Pathing-Based Trajectory with Gate Open

OUTREACH HIGHLIGHTS

- Hosted all SoCal League E meets at our School
- Mentored/collaborated with 25+ FTC Teams, 2 FRC Teams, 2 VRC teams
- Connected with Marine Biology Researcher, Paleontologist, Harvey College engineering and CS Professors, and Engineers at JPL
- Invited to the Western Association of Vertebrate Paleontology
- Introduced a team record number of people to FIRST

SEASON HIGHLIGHTS

- Highest OPR in SoCal League E
- 12 artifact 40pts Auto
- ~50 artifact Teleop
- ILT 4 Winning Alliance
- ILT 4 Inspire 3rd Place
- Regional Think 3rd Place
- Worlds Qual!!

SEASON GOALS

1 STAY WITHIN OUR PREALLOCATED ROBOT CONSTRUCTION BUDGET OF $5,000

OUR JOURNEY

Set **a hard $5,000 ceiling** after rebuilding from **major roster loss** in 2024-25

Utilized Excel & custom software for inventory cataloging and expense tracking.

Hosted **team discussions** on how to optimize robot utilizing available tools

Made **sacrifices** like using **MDF over machined parts** to reduce cost and risk during iterations, **added gusseting** to help **offset the durability trade-off**.

Spent less than our $5,000 budget and is able to invest remaining funds into next year's outreach and tool expenses.

2 ACHIEVE A SUB 7 SECOND CYCLE TIME FOR THE ROBOT'S SHOOTER

OUR JOURNEY

Calculated & identified sub-7-second cycle time as the threshold for playoff-level scoring

Modeled the mechanism in CAD, then validated with a cardboard prototype **before cutting materials**

Applied torque, projectile motion, and energy transfer calculations to optimize flywheel speed and launch angle

Iterated through **multiple revisions**, testing on a practice field and eliminating transfer bottlenecks

Created an fast and efficient mechanism that has a sub 5.7 second cycle, allowing our team to score high in playoffs

3 LESSEN DRIVER MENTAL AND PHYSICAL LOAD DURING MATCHES

OUR JOURNEY

Defined the goal — offload aiming, positioning, and repetitive actions from driver to software

Researched odometry and sensor fusion, selecting an odo pod setup matched to our drivetrain

Built an auto-aim system using math-based transfer angle modeling, **validated with hundreds hours of testing** and PID tuning

Studied open-source frameworks, then built **our own control pipeline** (60+ hours of integration) tailored to our hardware

Delivered a 1-driver competition design — automation handles aiming and flywheel, improving scoring consistency

4 SUCCEED IN JUNIOR-SENIOR TEAM TRANSITION PHASE & PREPARE NEXT PHASE

OUR JOURNEY

Learned from experience — losing most of our roster after 2024-25 showed us what knowledge loss does to a program

Paired seniors with junior mentees across every subsystem for weekly hands-on sessions

Documented critical systems — assembly guides, software architecture, wiring standards, and decision-making processes

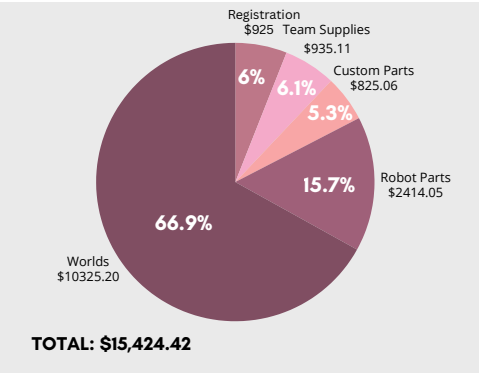
Shifted ownership progressively — juniors went from support roles to leading subsystem decisions by mid-season

Built a 2026-27 onboarding plan so Webb.EXE grows regardless of roster turnover

Sustainability Plan

Part of our mission is to **inspire the next generation of engineers through FIRST.** To bring new members into the program, we engage our school community through **open houses, admitted student events and tours, and scrimmages** where newcomers can **try robotics firsthand**. Our school is home to **two other FTC teams — 19888** Alfatech and **358** Gaulbots — and we love working alongside them, sharing knowledge and supporting each other whether that's helping with a mechanism, debugging code, or preparing for competition. When students from these teams are ready to take on a greater challenge, they **naturally find their way to us**, and we are always excited to welcome them. This connection between our teams creates a **supportive, tight-knit robotics community** where everyone grows together and keeps the team powered by a group of **passionate students**.

BUDGET DISTRIBUTION



TOTAL: $15,424.42

Financial Sustainability

- As **a school-based team**, we are in the privileged position of receiving stable institutional funding through **The Webb Schools**, which **covers our baseline operational needs**.
- **In addition to our school funding**, we are supported by the **generous sponsors** pictured to the left, who **contribute additional resources** ranging from **financial donations to in-kind support**. We are proud that many of these relationships were **personally initiated** by our student members — **from writing emails to presenting our team's mission** — **teaching** our students invaluable **real-world skills** in **communication and relationships.**
- To ensure **accountability** and **long-term sustainability**, we **track all expenses** in a shared budget spreadsheet, logging every purchase. This is reviewed regularly by **our treasurer** to keep our spending **aligned with our seasonal goals** and ensure we remain **financially healthy** year to year.

WEBB goBILDA ROCKBROS

fabworks. FOURIER Vision Express

TEAM STRUCTURE

This season marked a **significant transition** — most of our prior roster graduated, **leaving only 2 members** as returning members. **2 other members**, drawing on their **prior experience in robotics in FTC 19888**, stepped up to **lead their respective areas**. Our leadership structure includes **Captains and specialized leads** in software, build, and outreach, yet **remains flexible**, with **members contributing across multiple roles**. Through **mentorship**, experienced members guided newer teammates in both technical skills and team values, ensuring **every member grows as a leader** and **sustains our team year after year**.

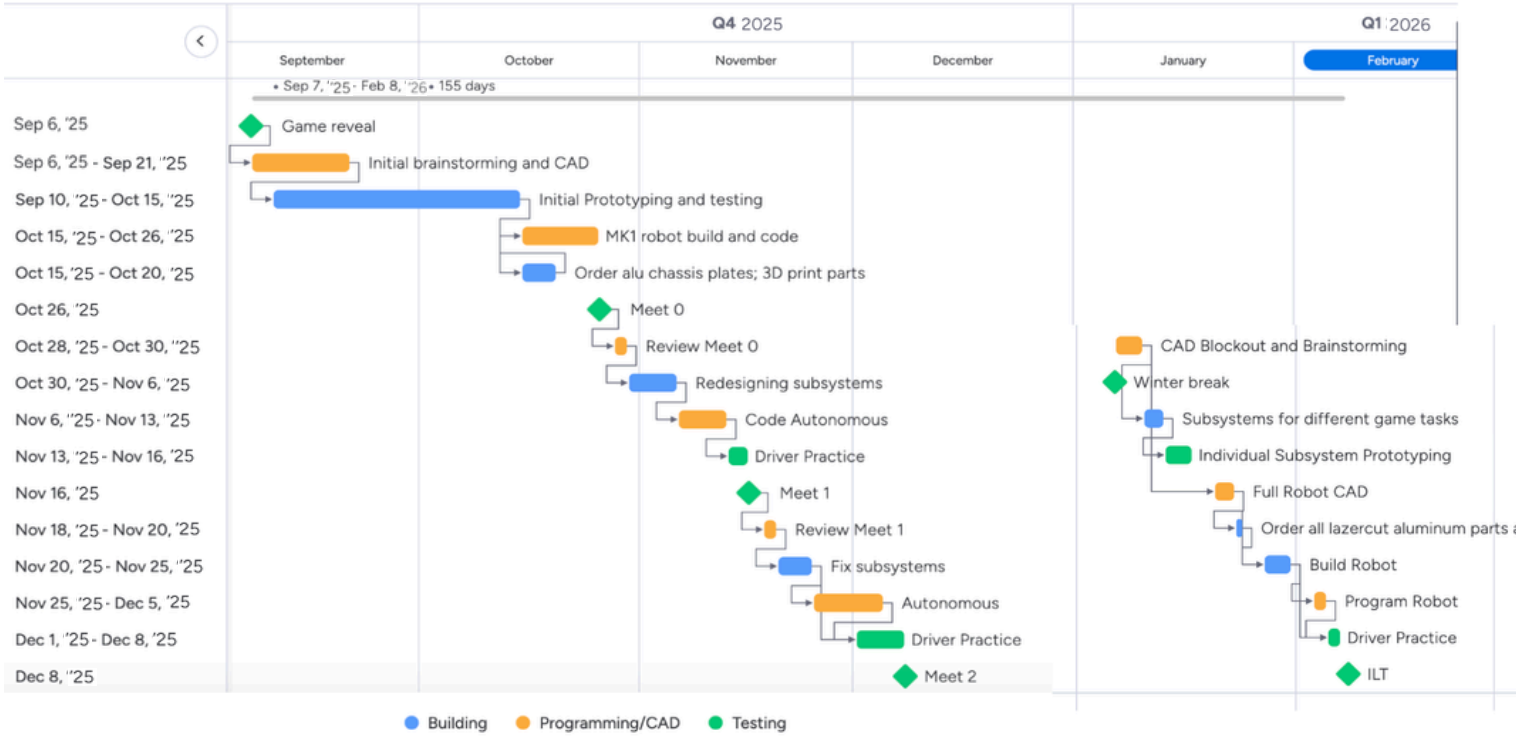
TEAM SKILLBUILDING

At the start of the robotics season, new members rotated through a variety of **team tasks** so that everyone could gain exposure to the full range of **skills** required in robotics. This system introduced members to different **team roles**, encouraged them to step outside their comfort zones, and helped them understand how each part of the team contributes to overall success. The process was guided by **veteran members**, who provided hands-on instruction and support throughout the season. After this rotation, members were assigned to positions that best aligned with their **interests** and **strengths**. Although not everyone could receive their preferred role, due to the need to fill essential **team responsibilities**, each member gained valuable experience and developed new abilities. Through the guidance of veterans and occasional support from other teams, we built a more **skilled**, **collaborative**, and **well-rounded** crew. In addition, our team maintains a strong culture of **mutual support** across all areas of work. When one subteam needs assistance, other members step in to help wherever needed. For example, when the mechanical team requires extra hands, members **readily contribute their time and effort**. Similarly, when the CAD team is evaluating design options, teammates **collaborate** through research, discussion, and feedback to help identify the most effective solution. This willingness to support one another **strengthens** both our **teamwork** and **overall efficiency**.

WORKFLOW

To stay organized, we use **a combination of communication and project management tools** — Discord and Instagram for team messaging and coordination, and Trello for tracking tasks and deadlines. However, **multiple platforms can create friction**; we are currently building our own **internal dashboard** that unifies these tools into a single **centralized workspace** where information is more **context-rich** and easy to act on. Once complete, we plan to **open-source** it and **share** it with the **broader FTC community**, giving other teams the same organizational foundation we benefit from.

PROGRESS TIMELINE



OUTREACH

Our team prioritizes outreach as a key way to build connections and **give back to communities in the FIRST way**. We've reached out to teams and professionals, hosted events of varying scales and purposes, and worked to reach a diverse range of audiences. This year, our theme was **Better Together** – **sharing our passion for engineering while making a positive**, lasting impact on science and engineering communities and **introducing FIRST**.

- **400+ IN-PERSON OUTREACH HOURS**
- **100K+ INFLUENCED**
- **20+ EVENTS**

MENTORS

Ms. Bailey Jorgensen

- Marine Biology researcher from UC Santa Barbara
- Talked to us about marine biodiversity and its relationship to environmental protection
- Collaborated with us on the fossil game for museum publicity and paleontology education for younger kids.

Robotics Professor Duan Yi

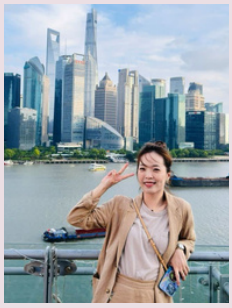
- Chinese Professor of robotics and engineering from BUAA
- Conversed and exchanged ideas regarding design principles

Mr. Hamilton

- FIRST Coach for 15+ years combining involvements in different program
- Taught us team management and engineering practices from his interaction with other prestigious robotics programs in SoCal and Canada

Lea Gan

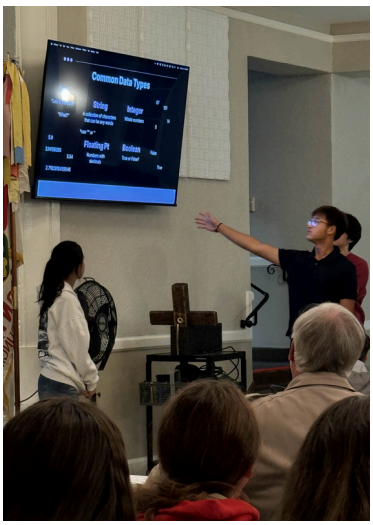
- Vice President of Fourier – robotics company developing rehabilitation and humanoid bots
- Introduced us to the progress and technology of the company
- Provided insight on the real life application of robotics technology

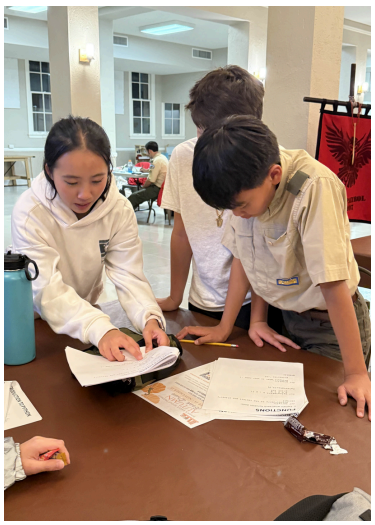


This year, we've collaborated with several professionals to leverage their knowledge in support of our development. We are immensely grateful for their guidance over the season, which has been instrumental in helping us reach our achievements so far.

Starting Local

As the **veteran team** in our school's robotics program, we place great value on team growth and development throughout the season. We take pride in **managing recruitment and training** for our school's robotics program, dedicating significant effort each year to **bring 30+ new students to campus** into the FIRST community and guiding them along the way. We also serve as the **technical backbone of our school's introductory engineering and computer science courses.** Through our commitment to mentorship, empowerment, and fostering a collaborative environment, we aim to inspire the next generation of innovators and problem-solvers within our school and the broader robotics community.





OUTREACH TIMELINE

10/14/2025 - **Local** Outreach **60+** influenced

Instructing Boy Scouts Python

- Taught introductory Python concept to Boy Scouts through hands-on activities
- Created a PowerPoint lesson and guided worksheet for structured learning
- Reinforced engagement with games and rewards to encourage probleming-solving

10/21/25 - **Local** Outreach **30+** influenced

Jet Propulsion Laboratory Tour

- Visited NASA Jet Propulsion Laboratory (JPL) headquarters for an educational tour
- Learned about real-world engineering applications, space missions, and robotics
- Gained insight into STEM careers and large-scale collaborative projects

11/2/25 - **Local** Outreach **100+** influenced

Discover Webb Day

- Hosted robotics, math, and science stations
- Showcased past robots and current season progress

11/20/2025 - **Local** Connection

Online Meeting with 7767 Blueprints

- Mentored autonomous programming strategy development
- Support in debugging autonomous and refined structure

November 2025 - **International** Outreach

Empowering North Korean Refugee Students

- Provided basic Python lessons to North Korean refugee students through bi-weekly and in-person sessions
- Gradually introduced core concepts including variables, conditionals, randomness, user input, and output formatting through interactive games
- Emphasized a supportive, inclusive, and flexible environment to express confidence and self expression

```
import random
import time

def play_바위_보_가위():
    choices = ["바위", "보", "가위"]
```

11/20/2025 - **Local** Outreach

Online Meeting with 7767 Blueprints

- Mentored autonomous programming strategy development
- Support in debugging autonomous and refined structure

11/23/2025 - **Local** Outreach **100+** influenced

Paleo Video Game

- Collaborated with the Raymond M. Alf Museum of Paleontology to develop an education mobile game featuring Baby Joe, a famous *Parasaurolophus*
- *Designed interactive gameplay where users make decisions about diet, habitat, migration, and predator avoidance*
- *Used the game to teach dinosaur biology, environmental challenges, and scientific thinking in an accessible format*

October 2025 - **Local** Outreach

Partnership with UCA

- Partnered with United Christian Academy (UCA) to launch a brand-new FIRST Tech Challenge (FTC) team at the middle school/high school level
- Collaborated directly with staff and students to introduce hands-on robotics and STEM learning experiences
- Expanded access to FIRST by recruiting new teams, coaches, and students with no prior exposure to the program

1/15/2026 - **Local** Connection

Meet with Blueprints

- Mentored autonomous strategy development and provided feedback on overall robot design
- Refined robot design, gave advice on structural improvements, and debugged code

2/21/2026 - **Regional** Outreach **100+** influenced

Fossil Fest

- Presented our educational game at Fossil Fest, engaging young children and families in interactive STEM learning
- Connected with industry professionals, introducing FIRST and building potential relationships with future mentors, sponsors, and advocates

3/20/2026 **International** Outreach

Admitted Students Day

- Showcased our robot introducing incoming students to our team and FIRST
- Engaged prospective students in STEM and robotics
- Strengthened long-term team growth by driving recruitment and expanding our future member pipeline

More Future Events Planned!

CREATING AN IMPACT

SEASON GOALS

- Make STEM more accessible
- Giving back to FIRST community
- Connecting with the local region
- Skills Development
- Mentoring Teams
- Archeology awareness

Alf Museum Game

January 21, 2026

In collaboration with the Raymond M. Alf Museum of Paleontology, our team developed **an educational game** centered on Baby Joe, a young *Parasaurolophus*. The game allows players to guide Baby Joe through key stages of growth and survival, making decisions about food, habitat, migration, and predator avoidance. The project introduces users to dinosaur biology, **environmental challenges, and scientific thinking** in an engaging format to be **widely accessible**.

Bridging Barriers in STEM

October, 2025 Onward

Our team had the incredible opportunity to collaborate with the Claremont community church. We connected with local **Boy Scout and Girl Scout organizations** to offer introductory programming courses, given **their challenges in accessing electronics and the internet.** We designed custom course materials and interactive worksheets to spread the love of engineering to more.

STEM for the Vulnerable

November, 2025 Onward

Following our programs supporting the local underresourced community, we identified the needs of vulnerable groups and the barriers they face to accessing STEM resources. This year, we've reached out to **refugees**, **victims of political controversies**. We established an interactive course for kids in need and opened up new possibilities by **providing long-term educational support**.

Submarine Project

March, 2024-Present

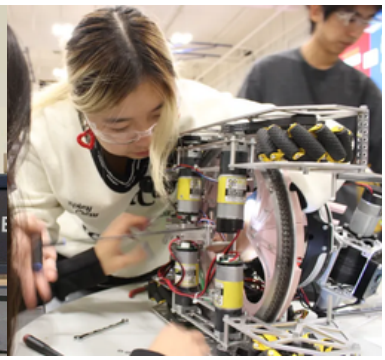
We initiated a **collaboration with the Raymond M. Alf Museum of Paleontology** on an **underwater fossil excavation project in Lake Skinner.** Through an extensive R&D process, we developed a **manually controlled submarine capable of underwater mapping and fossil detection**. The project was then integrated into our school's unique Unbounded Learning program and has become a long-standing project and selling point that combines our paleontology program with the Webb engineering experience. Extensions of this program are still ongoing to strengthen the interdisciplinary experience.

Community Connections

Our team is committed to expanding the robotics program within our community. To further develop our engineering and programming skills, we visited the nearby Claremont Colleges. We also connected with the Webb Admissions Department and engaged with **20+ prospective students**. During this time, we shared our past season's engineering portfolio and game footage to **promote FTC to those interested in joining the community.** Additionally, we hosted robotics, math, and science stations at open house events, where we spoke with over **100 prospective families** about the aforementioned programs.

League Connections

We have a longstanding tradition of hosting the Southern California League E. This year, we **hosted Meets 0, 1, 2, and 3 at the Webb Schools**. In addition to handling setup, cleanup, parking management, refereeing, and serving as MCs, we also actively engaged with our local community. We **participated in scrimmages and invited local teams to visit our lab**, offering them the opportunity to practice and learn from our engineering processes. Also **providing us with the unique opportunity to showcase FTC to our school.** We mentored newer teams during these events to guidance on strategy, design, and programming.



CREATING CONNECTIONS

MENTORING

As the veteran team in our school's 3-team system, we've all been through and value the development of rookie teams. In our school, we recruited members of all rookie teams and mentored them along the way. As part of giving back to the community, we are committed to mentoring and starting teams in FTC and VRC. Globally, we are active on social media and in online FIRST communities, and we provide technical support to teams in China, Canada, Bangladesh, and more.

FTC 358 GaulBots & FTC 19888 Alfa-Tech

- School sibling team we've recruited and mentored along the season
- Taught CAD, Prototyping practices and Programming

FTC 27353 Blue Lancers

- Helped integrade FTC dashboard tuning
- Assisted in flywheel PID loop tuning
- Introduced Megatag and Kalman filter for vision relocalization

FTC 7767 Blueprints

- Exchanged approach in advanced CAD
- Donated odometry pods and taught Road Runner

FTC 23929 Grizzly Tech

- Donated Parts
- Mentored on autonomous pathing.
- Donated Extra Control Hub during Competition

FTC 13368 Mechatronics

- Suggested professional engineering processes
- Supported team in CAD and Programming
- Shared Competition experiences

VRC 7700V Rolling Robotics

- Helped with the interview and designing
- Aided in Autonomous driving Tuning

FTC 13368 Mechatronics

- Suggested professional engineering processes
- Supported team in CAD and Programming
- Shared Competition experiences

VRC 2289A Shooting Stars

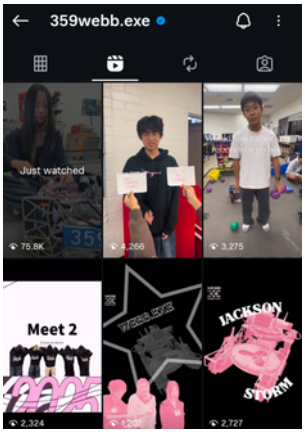
- Helped developed PID library for autonomous
- Mentored members for CAD design & Programming
- Assisted in VRC Worlds Qual

VRC 7700V Rolling Robotics

- Helped with the interview and designing
- Aided in Autonomous driving Tuning

Social Media Presence

This year, we also focused on posting across platforms to increase our team's presence and online influence in the local community, Across platforms we have reached over **239k views**, introducing stem to new communities and **raising FIRST awareness.**



CONNECTING

Aiming to form strong relationships and make presence in the FIRST community, we invited many teams to online and in person team meetings in order to exchange season experiences. Through these meetings we've built robust connections and co-hosted bigger outreach evens with these teams.

CONNECTED AND CONVERSED

FTC 15534 VERTEX
FTC 19600 Treeman
FTC 7767 Blueprints
FTC 26504 Axle-lotls
FTC 32480 Makers Assemble
FTC 6371 Machine Kings
FTC 23929 Grizzly Tech
FTC 27353 Blue Lancers

UPCOMING EVENTS

Our season does not end when competition ends. Despite our team all committed for States Championship preparation, we have already started to plan many upcoming outreach events for the off seasons.

- Official International Parts Sharing Platform for Environmental Protection
- Bike rides for marine garbage awareness
- Hackathons in China and Bangladesh
- Online Seminars for the local Community with industry professionals
- Engineering and Coding courses with local Non-Profits
- Women in STEM events with Harvey Mudd College CS Professor
- Collaboration with the education sector in Bangladesh
- Starting FTC teams in rural China

OUR DESIGN PROCESS

Having a planned design process is crucial for efficiently designing a high-performance robot. It allows us to systematically approach our design while considering all necessary factors and constraints. Under our unique team structure, each engineering decision is a collaborative result that considered the ideas of every member on the team.

1. Define the Problem
After developing a thorough understanding of the game, we define the task that the part or subsystem must accomplish.

2. Specify Requirements.
The objective of this step is to identify and define all constraints comprehensively. This ensures that potential issues during the implementation phase are minimized or avoided altogether.
- Are there any size or weight limitations for this part/subsystem?
- If applicable, how many motors or servos can it use?

3. Brainstorm ideas
Look back at previous seasons for potential inspirations on similar tasks, such as intakes. Research real-world problems that are similar and their solutions. Conducting research on real-world problems and their solutions, such as car suspension systems or crane arms, can provide valuable insights to optimize the robot's design and functionality for the current robotics competition.

4. Develop a preliminary design and quickly fabricate
Nothing fancy is needed in CAD besides a part that can be quickly manufactured using processes like laser cutting and 3d printing or built from standard parts. The prototype's purpose is not to be final but to enable us to test and determine its viability before committing further.

5. Test prototype
We test for:
- Effectiveness: does it solve the problem it was designed to?
- Weaknesses: what are the problems with the design?
- Practicality & Feasibility: similar to effectiveness. Is the design overcomplicated? Hard to manufacture?

6. Analyze Results
After testing, we reevaluate the design (how well can we optimize it?). Is it worth pursuing, or should we try a different approach?

7. Repeat from step 4 until we are ready to move on
We continue the prototyping and testing cycle until we feel that we have a design that we are confident in. We should have already improved it significantly from the initial prototype and should have learned enough from testing and analysis to make a final implementation.

8. Final Implementation in CAD
After improving the initial prototype and learning from testing and analysis, we can proceed with the final implementation. See the CAD organization page for how we keep track of our whole assembly.

SEPTEMBER 20, 2025

BREAKING DOWN THE GAME

We gather everyone to watch the kickoff video and read Game Manual Part 2 to understand the game and scoring tasks, spending half a day just to discuss the game and design strategy to make sure people are all on the same page.

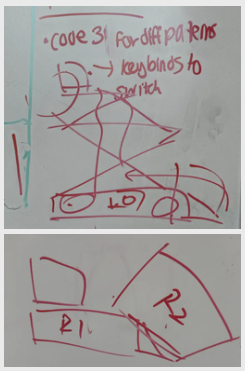
Key Functional Requirements (in order of R&D importance)
The robot can differentiate colors and effectively ingest artifacts in random positions. Additionally, the drive train speed and robot shooting speed should be as high as possible to reduce cycle time.

Subsystems required to meet Key Functional Requirements
Drivetrain, Tracking Turret, intake, Shooter mechanism

OCTOBER, 2025

BRAINSTORMING

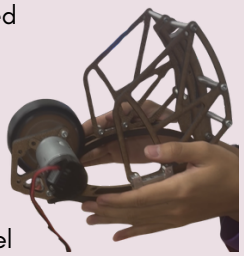
After understanding the game and design needs, we brainstormed, drawing from past robots, choosing a spindexer and kicker design with a flywheel shooter. We also explored intake options like a compliant wheel rapid fire and catapult.



NOVEMBER TO DECEMBER, 2025

PROTOTYPING

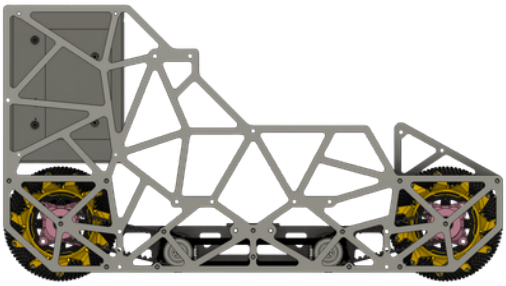
After brainstorming, we created CAD models of potential designs for testing in our next work session. Our focus was prototyping shooter mechanisms, as their performance would heavily impact other systems. Through testing, we selected a fly wheel shooter with adjustable hood. It proved more reliable, robust, and easier to integrate into our autonomous sequence, meeting our functional and design goals.



DRIVETRAIN

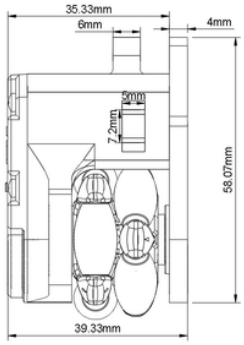
DRIVE TRAIN OVERVIEW

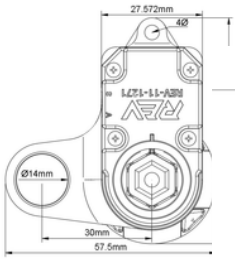
- The chassis features a custom parallel plate construction with a **belt-driven drivetrain** for **smooth and efficient motion**
- Equipped with 435 RPM motors and a 1:1 pulley ratio, the drivetrain achieves a **maximum speed of 0.696 m/s**
- The belt-drive system ensures smooth power transmission while minimizing maintenance requirements
- This chassis serves as the **foundational structure for key robotic modules**, including the intake, transfer, turret, and shooter
- To **mitigate backlash**, adjustable **belt tensioners** have been incorporated into the design
- We deliberately chose a low chassis design to accommodate 5in artifacts in different orientations, while adaptable to design changes.



Side View

Belt Ratio





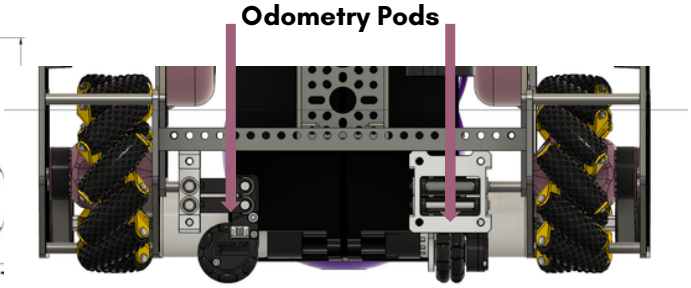
FUNCTIONAL REQUIREMENTS

- Integration of all subsystem mountings into 4 parallel plates, reducing excess weight.
- Consistent and adjustable belt tensioning to prevent backlash.
- Lightweight and aesthetically pleasing design for optimal performance and visual appeal.

ODOMETRY

Following our open-source innovation of the cost-effective odometry pods "TinyOdo" (Figure Left), **priced competitively at $65**, which provide high Clicks Per Rotation. We transitioned to utilizing the Gobilda 4-Bar odometry pods and the pinpoint odometry computer. The Pinpoint computer **offloads odometry calculations** from the main control hub, delivering **faster update rates** and more stable pose estimation, combined with our multi-threaded program. Handling encoder fusion and coordinate tracking in dedicated hardware has **elevated our autonomous capability** to a higher level.

Odometry Pods



V1 DESIGN

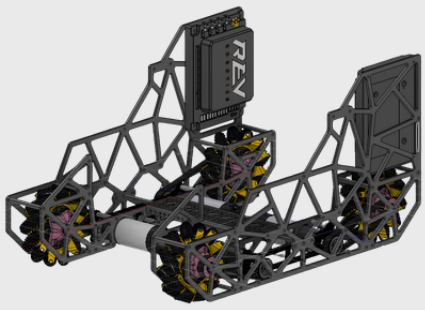
October 15, 2025

V1 Chassis

The previous belt driven system

V2 DESIGN

Jan 14, 2026

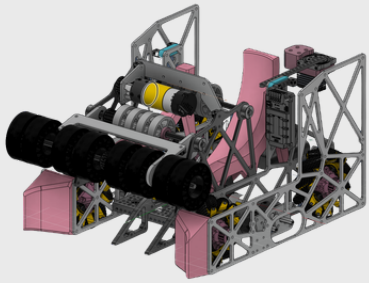


V2 Chassis

This version includes many new additions, including a redesigned belt and pulley system with improved tensioning for greater driving and autonomous accuracy. We also upgraded to the Pinpoint odometry system in this version.

FINAL DESIGN

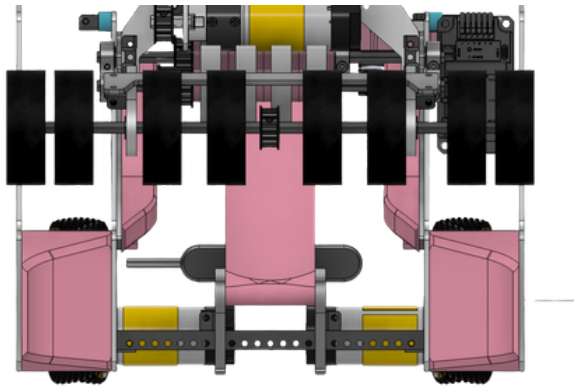
Feb 7, 2026



V3 Chassis

Our final version retains the same drive system, with V3 shortening the chassis to accommodate a full-width intake while maintaining rigidity. Frame rails were compacted for better mechanism packaging, enabling faster game-piece acquisition and reduced weight.

INTAKE

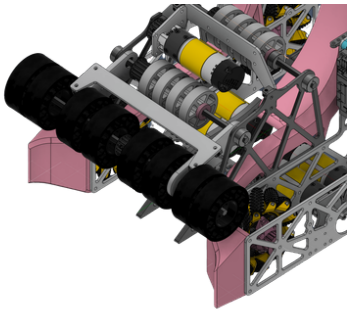


INTAKE OVERVIEW

- The intake was **redesigned** to span nearly **the full front width** of the robot, with rollers positioned further forward to **improve game-piece centering** and intake **consistency from multiple approach angles.**
- Angled vectoring bumpers were introduced to **passively guide** off-axis game pieces toward the center intake path, **reducing driver workload** and **increasing reliability** during high-speed cycles.
- The intake assembly mounts directly to reinforced frame nodes to ensure load transfer **without flex**, while the bumper structure maintains perimeter compliance and protects internal components.

FUNCTIONAL REQUIREMENTS

- Quick and efficient intake of artifacts from the ground.
- The capability to intake artifacts that were just released from the classifier.
- Lightweight and easy to maintain, with long-term reliability.
- Durability to withstand heavy use and frequent operation.
- Minimized time spent aligning and enhancing operational efficiency.
- Fast rotation speed to minimize intake time.
- Counter banded to lock artifacts inside chassis storage.



Low-Base Intake

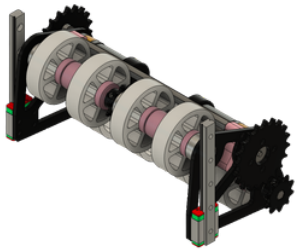
Facing inconsistencies with our previous iterations of an active intakes, we pivoted towards designing a mechanism it closer to the ground in verison 3. Our design utilizes a counter banded set of compliant wheels. By avoiding having to transfer artifacts from the ground up a high distance into robot storing area, our intake has became much more consistent and effective while demanding less torque on intake.

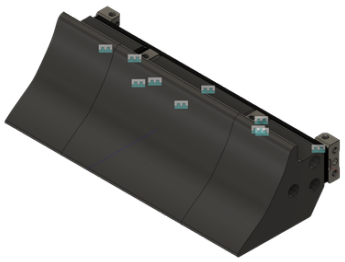
LEARNING PROCESS & PAST ITERATION

Our intake process has proven highly efficient for collecting game elements from the field. Earlier in the season, we experimented with a servo-powered intake on linear rails; however, we encountered challenges with optimizing servo power and intake speed. Consequently, we reverted to a larger, motor-powered intake with a passive alignment system that reliably transfers elements to our end effector, resulting in faster overall cycle times and higher reliability.

V1 DESIGN

September 17, 2025

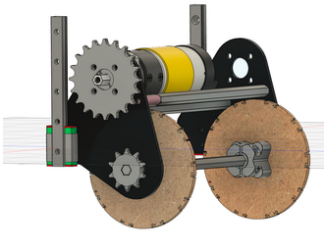


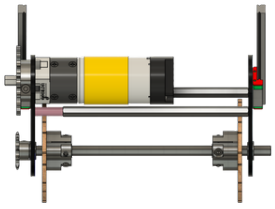


- 2 melonbotics servos.
- 2-stage vex compliant wheels.
- Counterstrung on a linear rail for vertical transfer.

V2 DESIGN

November 29, 2025

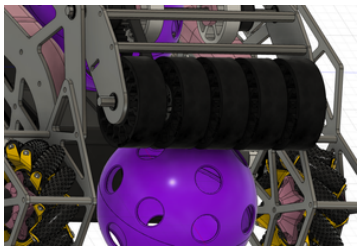


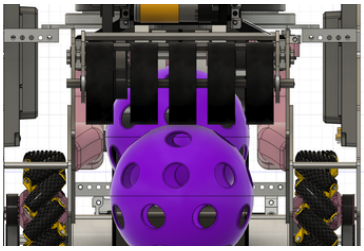


- Active intake design involving rubberbands.
- Fastest intake speed.
- 1:1 powered on a 435rpm motor.
- Issues with consistency and rubberbands breaking after cetain usage time.

V3 DESIGN

December 20, 2025



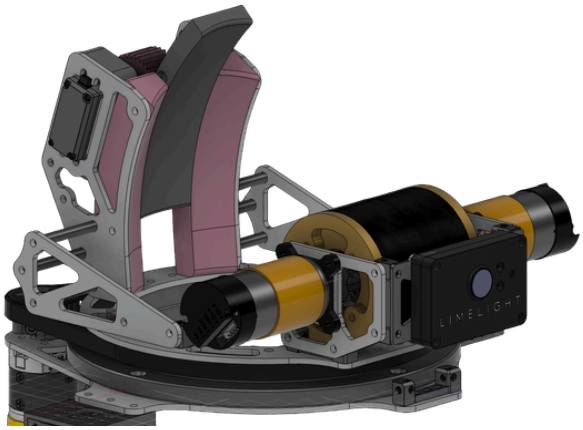


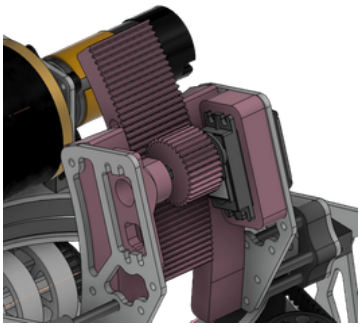
- Simpler, compliant wheel intake with a belt transfer.
- Alignment geometry from our initial prototyping.
- Allowed intaking from all orientations also serves as a lock for artifacts on the ramp.

SHOOTER

SHOOTER OVERVIEW

- Dual-motor direct-drive flywheel system replacing single-motor belt drive for improved responsiveness
- Flywheel weights added to **increase MOI**, reducing velocity drop between shots for **greater consistency**
- Retuned **compression geometry** to balance energy transfer and ramp-up time
- Redesigned **adjustable hood** provides precise launch angle control and more repeatable shot trajectories
- Low-profile architecture minimizes weight and preserves a compact center of mass





FUNCTIONAL REQUIREMENTS

- Deliver consistent moment of inertia cross variable distances and field positions
- Maintain a low vertical profile to reduce weight and mechanical complexity
- Support rapid consecutive shots without RPM recovery delays
- Interface with vision tracking for autonomous alignment and assisted teleop aiming
- Provide stable compression and contact time & adjustable angle for repeatable ball exit behavior

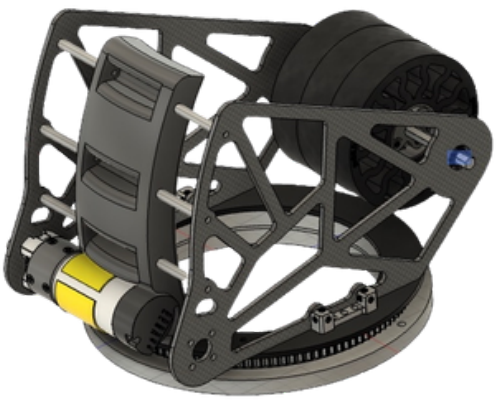
LEARNING PROCESS & PAST ITERATION

Through iteration, we found that a single-motor shooter introduced unnecessary losses and a slower spin-up time. Transitioning to a dual-motor direct-drive design significantly improved response time, torque delivery, and balance, especially during rapid-fire scoring. Extensive compression testing showed that small geometric changes had outsized effects on accuracy and ball behavior. The addition of an adjustable hood allowed us to decouple velocity from trajectory, while integrating Limelight 3A enabled closed-loop vision alignment without increasing shooter height. Overall, prioritizing a low-profile, efficient design resulted in a lighter, faster, and more consistent shooter system.

V1 DESIGN

October 28, 2025

The original design of the Shooter had one single 6k RPM motor on a non-adjustable hood. The shooter assembly is mounted on a lazy susan turntable, and the back is fully braced by a customized 3DP ramp. The turret that the shooter assembly sits on is powered by a servo belted to the turntable, allowing around 60 degrees of rotation while keeping the shooting angle constant.

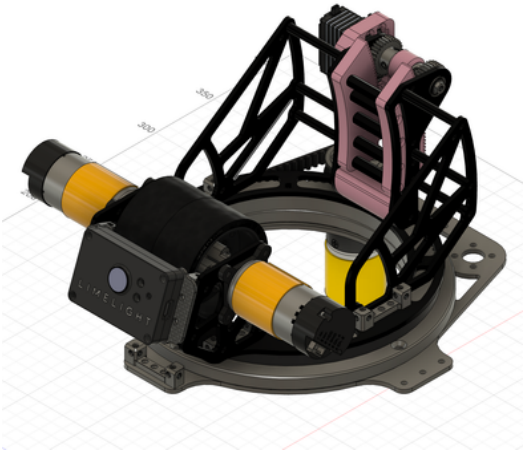


V2 DESIGN

January 16, 2026

The shooter used for ILT features is powered by 2 direct driven flywheels, improvements and iteration from V1 include:

- Made hood adjustable, regular to herringbone gear
- Tested and refined compression level to most optimal shooting power, balancing friction and stall force.
- Turret is now powered by a 435rpm motor for full freedom of rotation and smoother tracking response.



TRANSFER

TRANSFER OVERVIEW

- Redesigned around a continuous artifact path from intake to shooter, eliminating dead zones between stages
- Tilting linkage dynamically controls wheel compression to prevent jamming at high cycle speeds
- Improved alignment across all stages reduces sensitivity to artifact orientation for reliable high-speed cycling

TRANSFER GATE

- Automatic gate oscillates between idle and shooting states based on flywheel RPM
- Powered by one Axon Max and a goBILDA beam

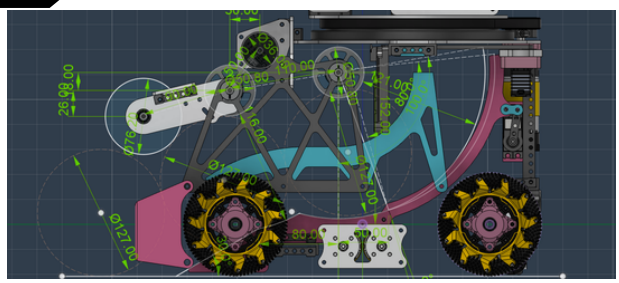
2-STAGE COMPLIANT WHEEL

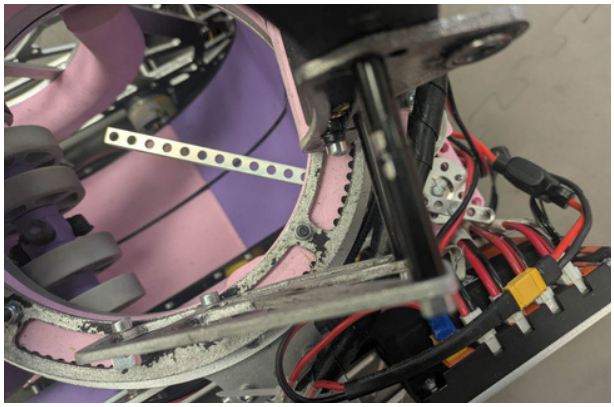
- Continuous transfer sequence, able to shoot 3 artifacts consecutively
- Improved over the scissor kicker for better stability
- 1:1 belted from a 1150 RPM motor with tested optimal compression

CUSTOM GEOMETRY RAMP

- Vectors artifacts toward the center of the robot
- Geometry tuned to naturally self-center artifacts, reducing jams and misfeeds
- Refined through iterative CAD and simulation testing to validate angles and clearances
- Fully integrated with the low-profile chassis maintaining a low center of gravity

FINAL ITERATION





FUNCTIONAL REQUIREMENTS

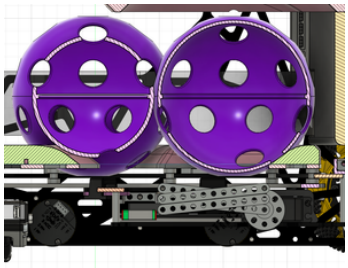
- Reliably intake and transfer artifacts from the front intake to the shooter without stalling or jamming
- Vector artifacts to the centerline of the robot for consistent shooter alignment
- Support continuous, rapid-fire scoring without requiring full system reset between shots
- Maintain compatibility with a low-profile chassis and preserve a low center of gravity
- Minimize cycle time between shots

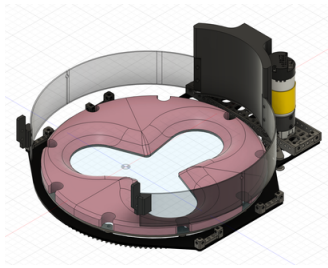
LEARNING PROCESS & PAST ITERATION

Through iterative testing and on-field trials, we learned that traditional kicker-based transfer systems significantly limit scoring speed and efficiency. Kickers can be mechanically complicated and introduce unavoidable delays due to retraction time, timing sensitivity, and inconsistent energy transfer between shots. In contrast, our continuous transfer and rapid-fire system enables a smooth, uninterrupted flow of artifacts, allowing multiple shots to be taken in quick succession without pausing or resetting the mechanism. This shift dramatically reduced cycle time, improved scoring consistency, and better matched the fast-paced nature of gameplay. As a result, we prioritized continuous motion, compliant wheel transfer, and automated gating over discrete kickers.

V1 DESIGN

September 30, 2025

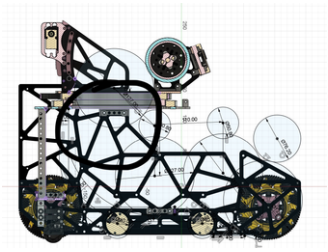


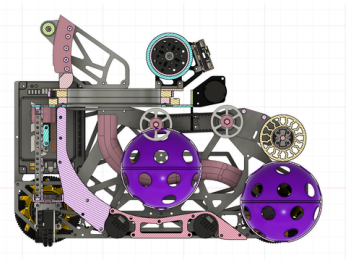


- The initial design was a direct driven scissor lift
- Single-action kicker used to push artifacts into the shooter one at a time
- Required full retraction and reset between shots, increasing cycle time
- Spindexer, requiring additional pid on position.
- Chain driven spindexing more prone to slack and jamming
- Limited throughput, making rapid multi-artifact scoring difficult
- More prone to desync under match stress or partial jams

V2 DESIGN

November 22, 2025





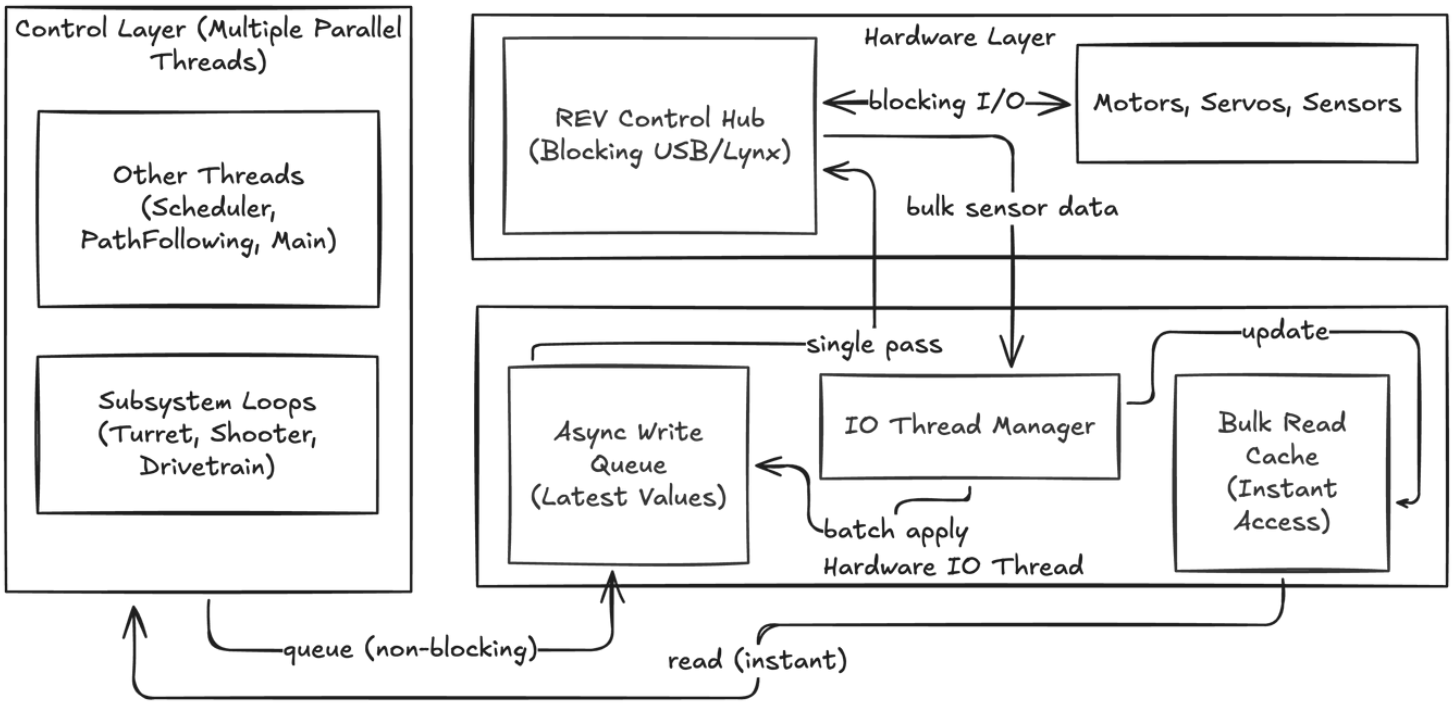
- Continuous, wheel-driven transfer maintains constant artifact motion
- Supports rapid-fire scoring with multiple artifacts fed consecutively
- Self-centering geometry vectors artifacts into the shooter consistently
- Eliminates reset delay, significantly reducing cycle time
- More tolerant to minor misalignment due to compliant wheels and a guided ramp
- Provides smoother, more repeatable shots under high-speed operation

TELEOP

TELEOP OVERVIEW

- Multithreaded Architecture: **6+ independent concurrent loops** (main, command schedule, IO thread, subsystem threads) run in parallel, **enabling simultaneous operation** without blocking
- Optimized Hardware IO: **Dedicated IO thread** performs **bulk sensor reads** and batches **motor/servo writes asynchronously**, eliminating **blocking delays** and **timing jitter** from **REV Control Hub USB transactions**
- Advanced Gamepad Control: **Non-linear exponential scaling** provides precise low-speed maneuvering with full power at high inputs, featuring **instant ~50Hz button response** for shooting, intake, turret control, and drivetrain
- Dual-Mode Operation: **Manual mode** for direct driver control, or **auto mode** with **zone-based shooting**, **turret lock-on**, and **dynamic vision pipeline** switching between AprilTag localization and **alliance-specific targeting.**

```
// DEDICATED IO THREAD - Bulk reads & batch writes
class IOThread: RobotThread("IOThread", 0) {
    override fun runLoop() {
        for (lynxModule in RobotHardware.lynxModules) {
            lynxModule.clearBulkCache()  // Bulk read
        }
        HardwareDevices.applyAllAsync()  // Batch apply
    }
}

// ASYNC INTERFACE - Non-blocking writes
motor.async { setPower(0.8) }  // Queues instantly

// Atomic swap prevents blocking
val toApply = pendingCalls.getAndSet(ConcurrentHashMap())
for ((method, args) in toApply) method.invoke(device, *args)

// CONCURRENT SUBSYSTEMS - 6+ independent loops
abstract class ThreadSubsystem(
    name: String,
    updateIntervalMs: Long = 20
) : RobotThread(name, updateIntervalMs)

// MULTITHREADED INITIALIZATION
threadManager.addThread(CommandScheduler())
threadManager.addThread(IOThread())
threadManager.addThread(PedroPathThread())
initOpMode()  // Subsystems auto-register
threadManager.startAll()  // All threads run in parallel
```



LEARNING PROCESS

We initially used a traditional single-threaded, sequential structure for our first league meet, but switched to multithreading before Meet 2 to eliminate lag when controlling multiple mechanisms simultaneously. After observing inconsistent timing and hardware delays, we researched FTC optimization strategies and consulted our alumni, leading us to implement a dedicated IO thread that performs bulk sensor reads and batches motor/servo writes asynchronously—dramatically improving control loop consistency. We also replaced our linear gamepad mapping with a non-linear exponential relationship, giving drivers finer control at low speeds while preserving full-speed responsiveness. The transition came with challenges including race conditions between commands accessing the same subsystem (solved with requirement-based scheduling) and synchronization issues where threads would run out of sync, causing unpredictable behavior that required careful coordination through thread-safe methods and volatile state variables.

```
// From MovementSubsystem.kt
private fun scaleInput(
input: Double,
exponent: Double,
linear: Double
): Double {
    if (input == 0.0) return 0.0
    return abs(input)
      .pow(exponent)
      .withSign(input) * linear
}

fun setDriveValues(
y: Double,
x: Double,
rotation: Double
) {
    // Forward/backward
    driveY = scaleInput(y, 1.8, 1.8)
    // Strafe
    driveX = scaleInput(x, 1.8, 1.8)
    // Turn (higher exponent = finer control)
    driveRot = scaleInput(rotation, 2.0, 1.2)
}
```

Autonomous Overview

The autonomous code for our robot is designed to optimize path planning and execution using **Pedro Pathing**, a tool that leverages **Bézier curves** for smoother and more efficient movement. This allows for better error correction and ease of implementation compared to traditional splines. Additionally, we incorporated our own **custom multi-threaded command-based system** to streamline complex sequences by replacing switch statements with path-driven automation.

Functional Requirements

- Implement Pedro Pathing for autonomous navigation using Bézier curves.
- Integrate our custom multi-threaded command-based system for command-based execution of sequential robot actions.
- Ensure efficient handling of shooting and intake system delays without requiring a finite-state machine.
- Improve pathing accuracy and speed while maintaining ease of use.
- Follows paths with maximum motor power to complete 12-ball auto with gate releases

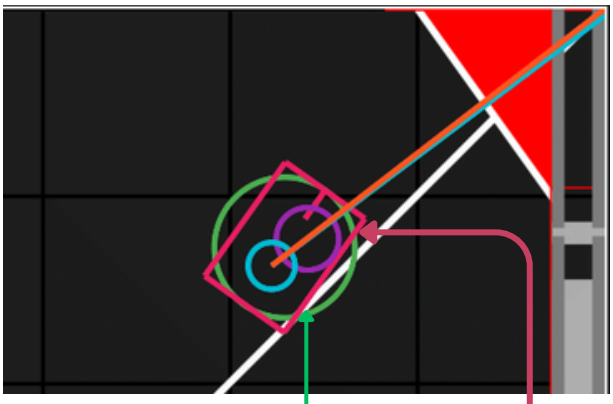
This is our second year using Pedro Pathing, and after gaining experience last season, we are confident it remains the right choice for our team. However, we did encounter some challenges. Unlike Road Runner, which uses time-based path execution, Pedro Pathing's distance-based approach occasionally caused paths to run longer than expected. To address this, we implemented path timeouts to ensure consistent cycle times. We also improved localization accuracy by fusing vision-based localization with Pinpoint dead wheel odometry, resulting in more reliable and precise autonomous routines.

Command Based Autonomous Routine

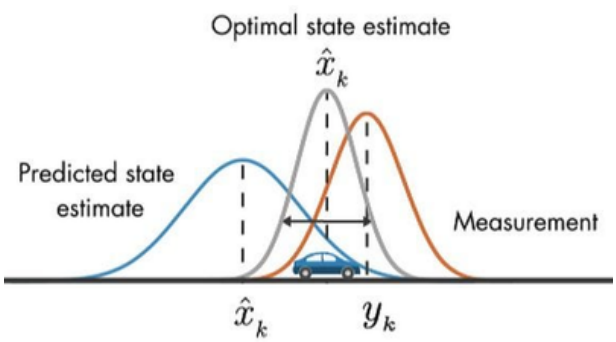
Our intake and transfer systems require a lot of time-based delays between actions, which cannot be implemented efficiently in a finite-state machine. As a result, we use a command-based system, which organizes all autonomous actions such as servo movements and path following into commands. These commands are called sequentially and in parallel using wrappers, which allows for complex simultaneous movements with multiple PIDF controllers and servo sequences.

```
SequentialCommandGroup.of(
    setCloseShooting(),

    PedroPathingCommand(startToShoot,0.7),
    shootBalls(),

    // Ball 3 cycle: pickup and shoot
    PedroPathingCommand(shootToBall3,0.9),
    intakeTransferOn(),
    PedroPathingCommand(ball3Intake,0.7),
    intakeTransferHalf(),
    PedroPathingCommand(ball3ToGate,0.7),
    PedroPathingCommand(gateToShoot,0.9),
    shootBalls(),

    // Ball 2 cycle: pickup and shoot
    PedroPathingCommand(shootToBall2,0.9),
    intakeTransferOn(),
    PedroPathingCommand(ball2Intake,0.7),
    intakeTransferHalf(),
    PedroPathingCommand(ball2ToGate, 0.65),
    PedroPathingCommand(gateToShoot, 0.9),
    shootBalls(),

    PedroPathingCommand(shootToBall1,0.9),
    intakeTransferOn(),
    PedroPathingCommand(ball1Intake,0.7),
    intakeTransferHalf(),
    PedroPathingCommand(ball1ToShoot,0.9),
    shootBalls(),

    // Park
    PedroPathingCommand(shootToPark)
)
```

```
class PedroPathingCommand(
val path: PathChain,
val holdEnd: Boolean? = null,
val power: Double? = null
): Command(false) {
    constructor(
        path: PathChain,
        power: Double
    ): this(path, true, power)

    constructor(
        path: Path
    ): this(PathChain(path), true, 1.0)

    init {
        require(!(power != null
        && holdEnd == null)) {
            "Speed parameter requires holdEnd"
        }
    }

    override fun initialize() {
        RobotHardware.withFollower {
            if (isBusy) {
                finishCommand()
                return
            }
            when {
                holdEnd != null
                && power != null
                -> followPath(path, power, holdEnd)
                holdEnd != null
                -> followPath(path, holdEnd)
                else -> followPath(path)
            }
        }
    }
}
```



Dead wheel odometry Vision Pose



Kalman Filter

A Kalman filter estimates the true state of a system by blending predictions with noisy measurements. It works in two phases: the predict step projects the state forward using a system model while growing uncertainty over time, and the update step incorporates new measurements. The Kalman gain determines how much to trust each source—if the measurement is precise relative to the prediction's uncertainty, the filter shifts toward it; otherwise, it favors the prediction. The result is a weighted estimate that minimizes overall error.

In our localization, dead wheel odometry drives the predict step—it's fast and smooth but drifts over time due to wheel slip and accumulated error. The Limelight provides the update step whenever it sees an AprilTag, giving us an absolute field position that doesn't drift but is less frequent and can be noisy. The Kalman filter combines both: we get the responsiveness of odometry with the drift correction of vision.

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